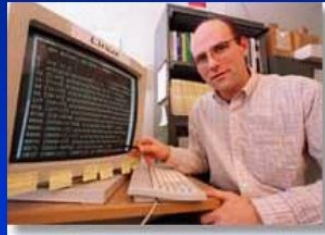


First Success: 1996

The theorem prover EQP (Equational Prover) achieved a significant milestone in AI history by solving the Robbins conjecture, a long-standing problem in algebra.

The success of EQP in solving the Robbins conjecture underscored the potential of AI to tackle complex mathematical problems and contributed to renewed interest and investment in AI research during the late 1990s and early 2000s.

1996: EQP proves that Robbin's Algebras are all boolean



----- EQP 0.9, June 1996 -----

The job began on eyas09.mcs.anl.gov, Wed Oct 2 12:25:37 1996
UNIT CONFLICT from 17666 and 2 at 678232.20 seconds.

----- PROOF -----

2 (wt=7) [] $\neg(n(x + y) = n(x))$.

3 (wt=13) [] $n(n(n(x) + y) + n(x + y)) = y$.

5 (wt=18) [para(3,3)] $n(n(n(x + y) + n(x) + y) + y) = n(x + y)$.

6 (wt=19) [para(3,3)] $n(n(n(n(x) + y) + x + y) + y) = n(n(x) + y)$.

.....

17666 (wt=33) [para(24,16426).demod([17547])] $n(n(n(x) + x) \dots$

[An Argonne lab program] has come up with a major mathematical proof that would have been called creative if a human had thought of it.

-New York Times, December, 1996

1997: Deep Blue

In 1997, IBM's Deep Blue made history by defeating Garry Kasparov, the reigning world chess champion, in a six-game match. This event was a landmark achievement in the field of artificial intelligence, demonstrating the potential of AI systems to perform at the highest levels in strategic games.



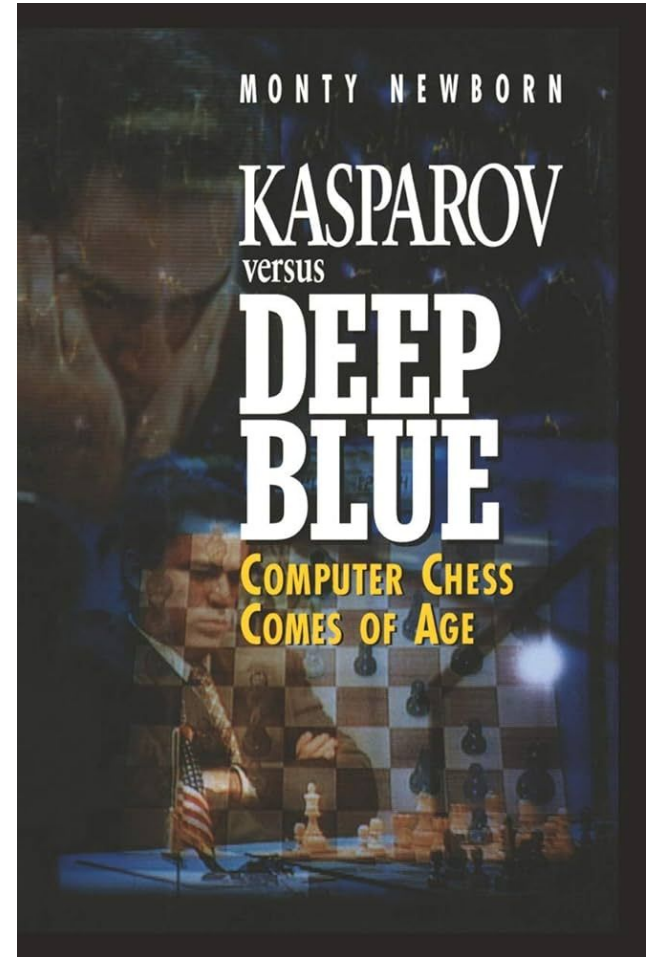
Success Story: Chess

Does Deep Blue uses AI?

“ If it works, its not AI”

Drew McDermott, a computer science professor at Yale:

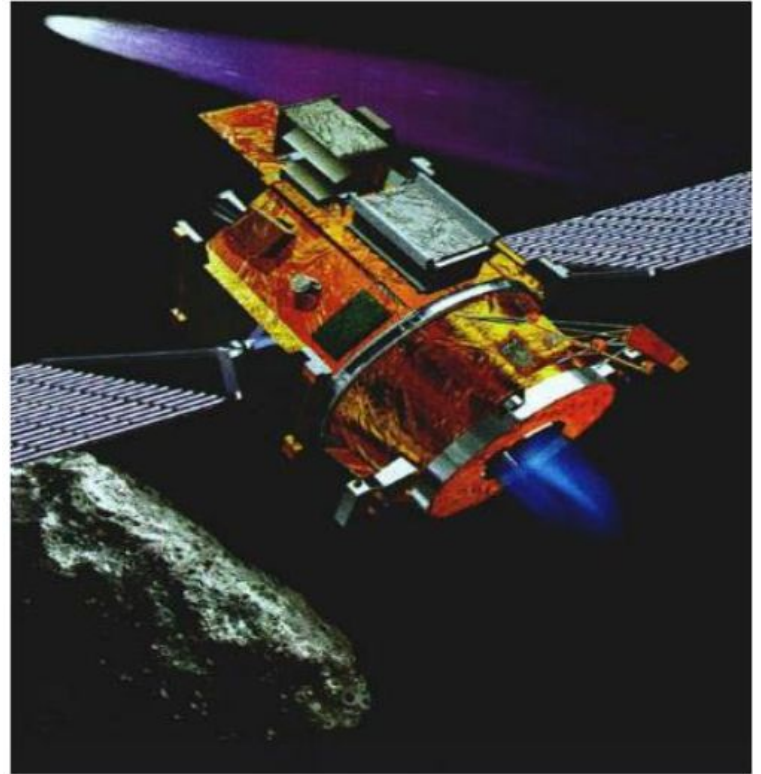
“Saying Deep Bleu doesn't really think about chess is like saying an airplane doesn't really fly because it doesn't flap its wings.”



1999: Deep Space I

In 1999, DS1 became the first spacecraft to use an AI-based autonomous system called the Remote Agent Experiment.

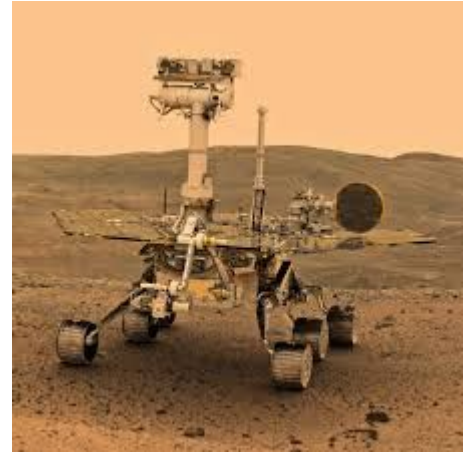
The RAE was capable of planning and executing its own activities, detecting and diagnosing problems, and making decisions without real-time human intervention.



2004 - 2009: Mars Rovers

NASA's Mars Exploration Rovers, Spirit and Opportunity, landed on Mars.

They utilized AI for autonomous navigation, allowing them to make decisions about which paths to take to avoid obstacles and choose scientifically interesting targets.



2005: Self-driving Cars

The 2005 DARPA Grand Challenge was a pivotal event in the development of self-driving car technology.

Organized by the Defense Advanced Research Projects Agency (DARPA), the challenge aimed to advance autonomous vehicle technology for potential military applications.

The success of the 2005 competition marked a significant milestone in the field of artificial intelligence and robotics.



2011: IBM's Watson

In 2011, IBM's Watson made history by competing on the quiz show "Jeopardy!" and defeating two of the show's greatest champions, Ken Jennings and Brad Rutter.

This event was a major milestone in the field of artificial intelligence, showcasing Watson's advanced natural language processing, information retrieval, and machine learning capabilities.

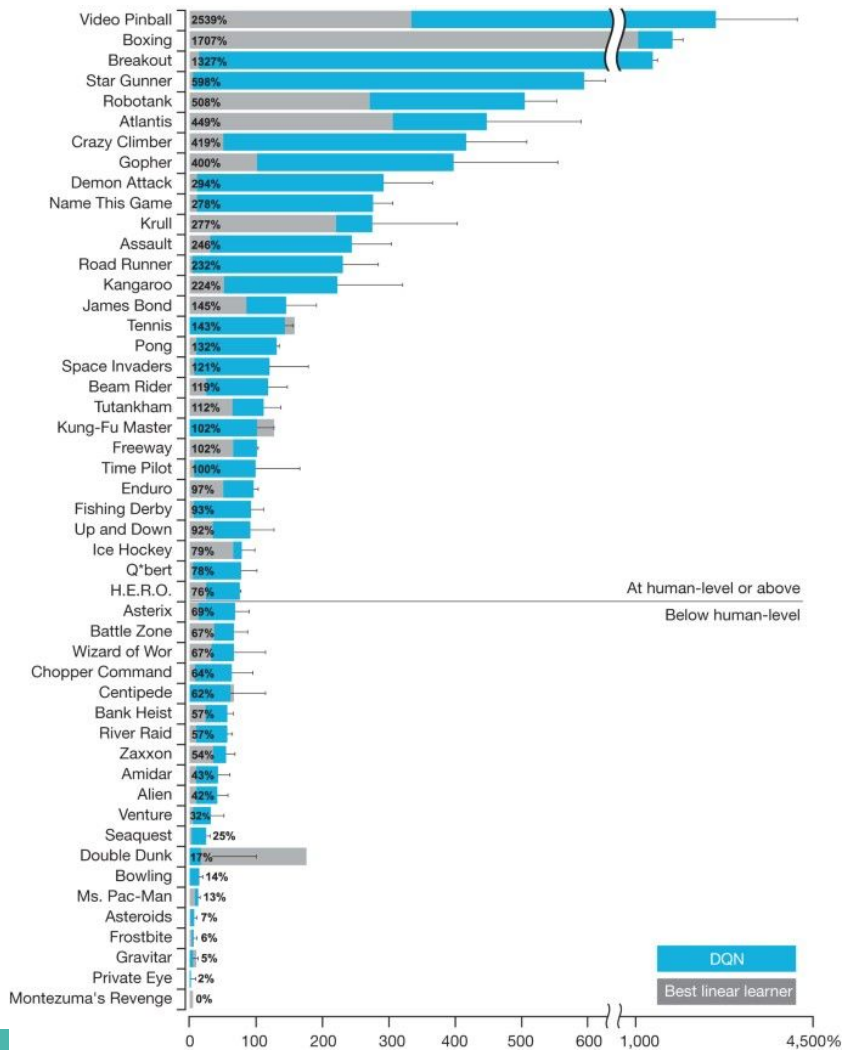


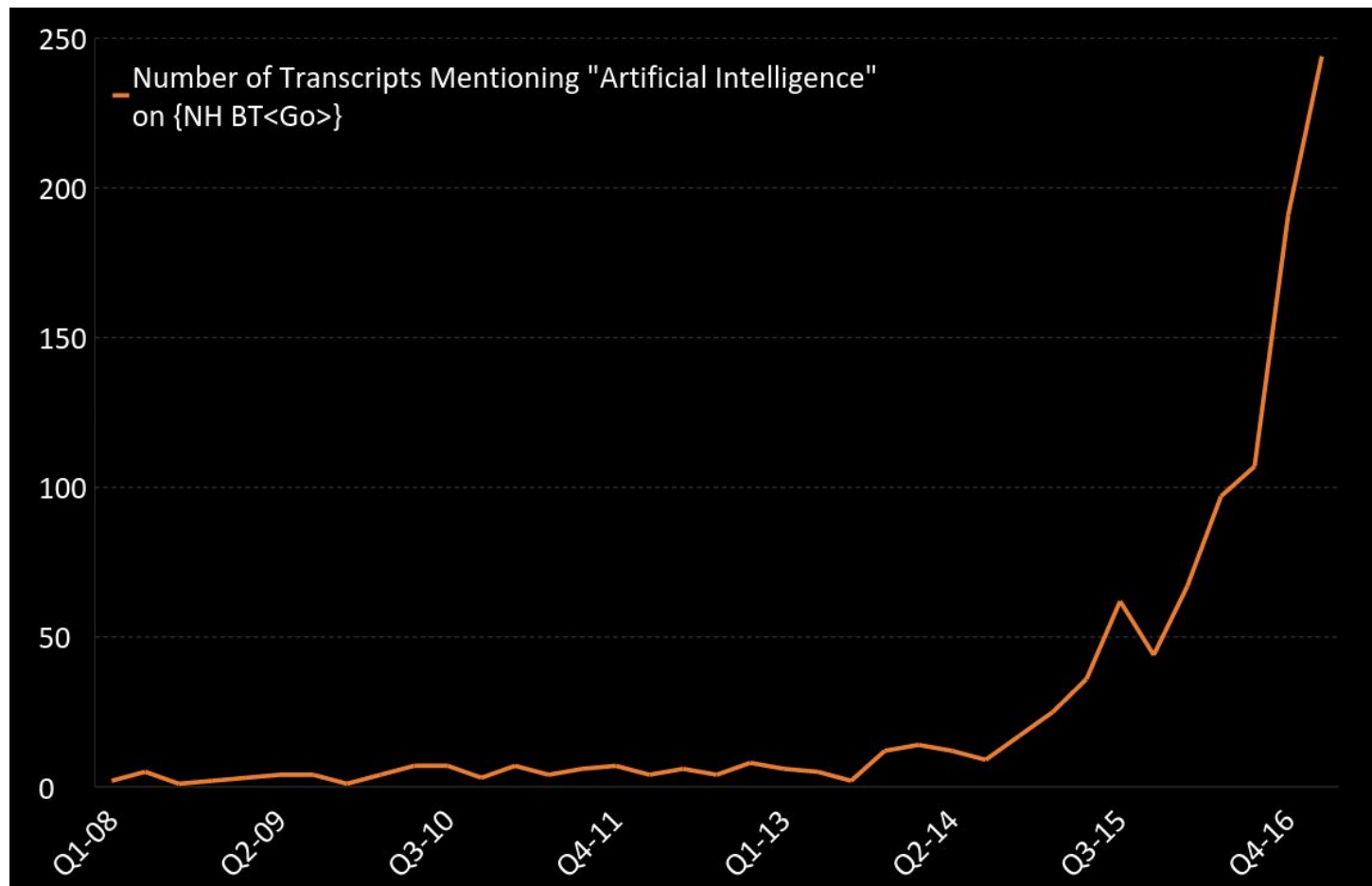
2016: AlphaGo

In 2016, Google's DeepMind created a groundbreaking AI program called AlphaGo, which made history by defeating one of the world's top Go players, Lee Sedol.



Human-level control through deep reinforcement learning





What Changed?

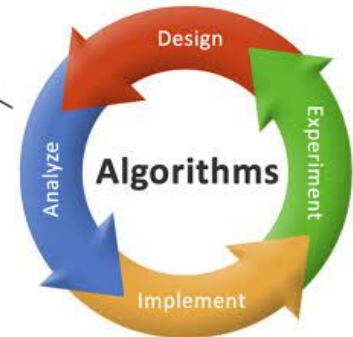


DATA

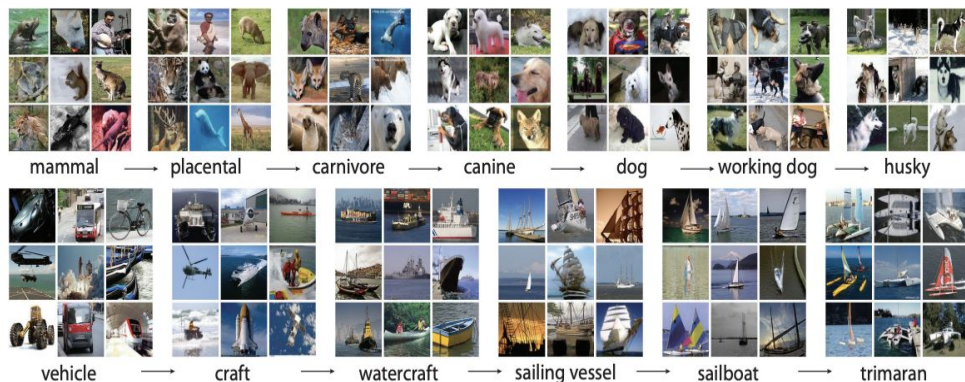
DEEP
NEURAL
NETWORKS

COMPUTATION
POWER

ALGORITHM



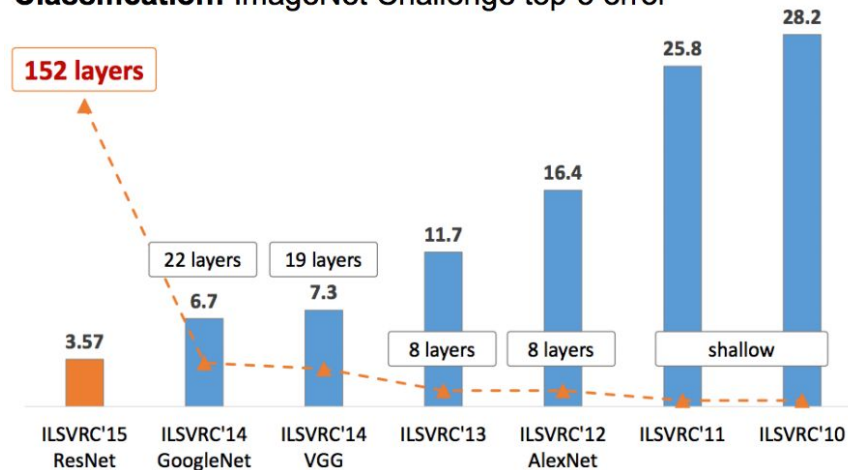
Object Recognition



CIFAR

CANADIAN INSTITUTE
for ADVANCED RESEARCH

Classification: ImageNet Challenge top-5 error



Artistic Applications

- Doodle to painting



- Style Transfer



- Image Colorization



Image to Caption

A person riding a motorcycle on a dirt road.



Two dogs play in the grass.



A skateboarder does a trick on a ramp.



A dog is jumping to catch a frisbee.



A group of young people playing a game of frisbee.



Two hockey players are fighting over the puck.



A little girl in a pink hat is blowing bubbles.



A refrigerator filled with lots of food and drinks.



A herd of elephants walking across a dry grass field.



A close up of a cat laying on a couch.



A red motorcycle parked on the side of the road.



A yellow school bus parked in a parking lot.



“If it works its not AI”



“It’s all AI”

Elon Musk predicts AI will overtake humans to the point that 'biological intelligence will be 1%'

Hype Or Reality: Will AI
Really Take Over Your
Job?

Robots TAKING OVER: AI to 'sink world into unemployed despair in hellish dystopia'

Robocup

By 2050, develop a team fully autonomous humanoid robots that can win against the human world champion team in soccer.



<https://www.youtube.com/watch?v=80qVUejnj8U>

What is AI?

Science of AI

Physics: Where did the physical universe come from?

And what law guide its dynamics?

Biology: How did biological life evolve?

And how do living organisms function?

Artificial Intelligence: What is the nature of intelligent thought?

What is Intelligence?

Dictionary.com: capacity for learning, reasoning, understanding, and similar forms of mental activity

Ability to perceive and act in the world

Reasoning: proving theorems and medical diagnosis

Planning: Take decisions

Learning and adaptation: recommend movies, learn traffic patterns

Understanding: Text, speech, visual scenes

Intelligence vs. Humans

Are humans intelligent?

- Replication human behaviour early hallmark of intelligence

Are humans always intelligent?

Can non-human behaviour be intelligent?

Some Definitions of AI

“ We call programs intelligent if they exhibit behaviors that would be regarded intelligent if they were exhibited by human beings. ”

— Herbert Simon

“ Physicists ask what kind of place this universe is and seek to characterize its behavior systematically. Biologists ask what it means for a physical system to be living. We in AI wonder what kind of information-processing system can ask such questions. ”

— Avron Barr and Edward Feigenbaum

Some Definitions of AI

“ AI is the study of techniques for solving exponentially hard problems in polynomial time by exploiting knowledge about the problem domain. ”

— Elaine Rich

“ AI is the study of mental faculties through the use of computational models. ”

— Eugene Charniak and Drew McDermott

What is Artificial Intelligence?

	human-like	vs.	rational
thought	“[automation of] activities that we associate with human thinking, activities such as decision making, problem solving, learning...” (Bellman 1978)		“The study of mental faculties through the use of computational models” (Charniak & McDermott 1985)
vs.			
behavior	“The study of how to make computers do things at which, at the moment, people are better” (Rich & Knight 1991)		“The branch of computer science that is concerned with the automation of intelligent behavior” (Luger & Stubblefield 1993)

What is Artificial Intelligence?

	human-like	vs.	rational
thought	Systems that think like humans		Systems that think rationally
vs.			
behavior	Systems that act like humans		Systems that act rationally

Thinking Humanly

- Cognitive Science
 - Very hard to understand how humans think
 - Post-facto rationalizations, irrationality of human thinking
- Do we want a machine that beats humans in chess or a machine that thinks like humans while beating humans in chess?
 - Deep Blue supposedly DOESN'T think like humans
- Thinking like humans important in Cognitive Science applications
 - Intelligent tutoring
 - Expressing emotions in interface.. HCI

Acting Rationally

- Rational behavior: doing the right thing
- Need not always be deliberative
 - Reflexive
- Aristotle (Nicomachean ethics)
 - Every art and every inquiry, and similarly every action and every pursuit is thought to aim at some good.

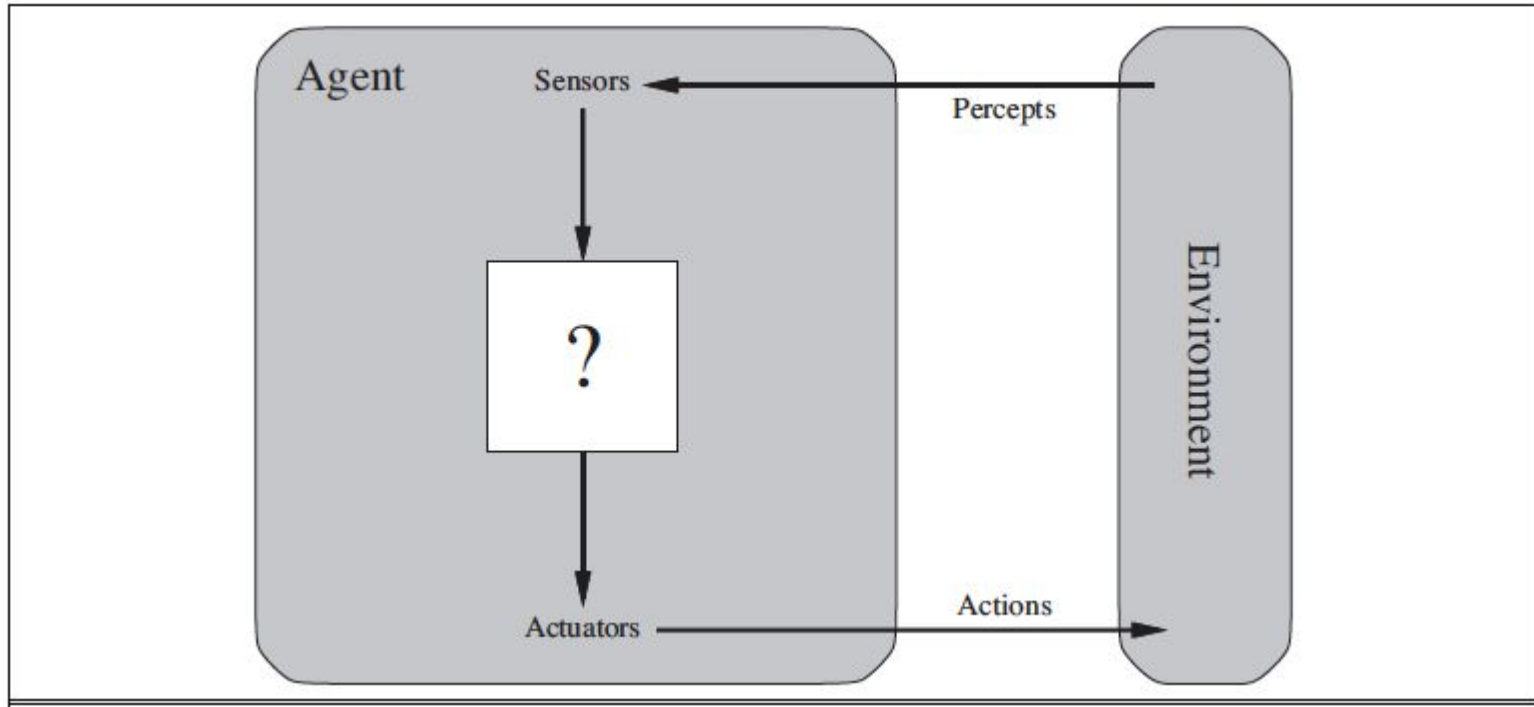
Acting Thinking?

- Weak AI hypothesis vs. Strong AI hypothesis
 - Weak Hyp: machines could act as if they are intelligent
 - Strong Hyp: machines that act intelligent have to think intelligently too

Intelligent Agents

Agent

- Formally “An **agent** is anything that can be viewed as perceiving its **environment** through **sensors** and acting upon that environment through **actuators**.”
- We use the term **percept** to refer to the agent’s perceptual inputs at any given instant.
- An agent’s **percept sequence** is the complete history of everything the agent has ever perceived.
- “In general, an agent’s choice of action at any given instant can depend on the entire percept sequence observed to date, but not on anything it hasn’t perceived.”
- An agent’s behavior is described by the **agent function** that maps any given percept sequence to an action.



Agents interact with environments through sensors and actuators.

PEAS (Performance, Environment, Actuator and Sensor) description

Agent Type	Performance Measure	Environment	Actuators	Sensors
Taxi driver	Safe, fast, legal, comfortable trip, maximize profits	Roads, other traffic, pedestrians, customers	Steering, accelerator, brake, signal, horn, display	Cameras, sonar, speedometer, GPS, odometer, accelerometer, engine sensors, keyboard

PEAS description of the task environment for an automated taxi

Agent Type	Performance Measure	Environment	Actuators	Sensors
Medical diagnosis system	Healthy patient, reduced costs	Patient, hospital, staff	Display of questions, tests, diagnoses, treatments, referrals	Keyboard entry of symptoms, findings, patient's answers
Satellite image analysis system	Correct image categorization	Downlink from orbiting satellite	Display of scene categorization	Color pixel arrays
Part-picking robot	Percentage of parts in correct bins	Conveyor belt with parts; bins	Jointed arm and hand	Camera, joint angle sensors
Refinery controller	Purity, yield, safety	Refinery, operators	Valves, pumps, heaters, displays	Temperature, pressure, chemical sensors
Interactive English tutor	Student's score on test	Set of students, testing agency	Display of exercises, suggestions, corrections	Keyboard entry

Examples of agent types and their PEAS descriptions.

Rational Agent

A **rational agent** is one that does the right thing. Obviously, doing the right thing is better than doing the wrong thing, but what does it mean to do the right thing?

Moral philosophy has developed several different notions of the “right thing,” but AI has generally stuck to one notion called **consequentialism**: we evaluate an agent’s behavior by its consequences.

This notion of desirability is captured by a **performance measure** that evaluates any given sequence of environment states.

Rationality

What is rational at any given time depends on four things:

- The performance measure that defines the criterion of success.
- The agent's prior knowledge of the environment.
- The actions that the agent can perform.
- The agent's percept sequence to date.

This leads to a definition of a **rational agent**:

"For each possible percept sequence, a rational agent should select an action that is expected to maximize its performance measure, given the evidence provided by the percept sequence and whatever built-in knowledge the agent has."

What is an Intelligent Agent?

An Intelligent Agent (IA) is

- a system that perceives its environment and takes actions to maximize its chances of success.
- These agents can be software-based, hardware-based, or a combination of both,
- and they operate autonomously or semi-autonomously to achieve specific goals.

Characteristics of Intelligent Agents

- **Autonomy:** Agents operate without direct human intervention and have control over their actions and internal state.
 - Self-Driving Cars
- **Perception:** They perceive their environment through sensors or data inputs.
 - Security Surveillance Systems
- **Action:** Agents take actions that affect their environment.
 - Robotic Vacuum Cleaners

Characteristics of Intelligent Agents

- **Rationality:** They act rationally to achieve their goals, which means they make decisions that maximize their expected utility.
 - Automated Trading Systems
- **Learning:** Some agents can improve their performance over time through learning from experiences.
 - Recommendation Systems

Agent Environment

- Definition: The environment in which agents operate.
- Examples:
 - Real-world environments, virtual environments, simulated environments.
- Types of Environments:
 - Fully Observable vs. Partially Observable
 - Deterministic vs. Stochastic
 - Episodic vs. Sequential
 - Static vs. Dynamic
 - Discrete vs. Continuous
 - Single-agent vs. Multi-agent
 - Known vs. Unknown

Fully Observable vs. Partially Observable

Fully Observable Environment:

- The agent has access to the complete state of the environment at each point in time.
- There is no hidden information, allowing the agent to make fully informed decisions.



Partially Observable Environment:

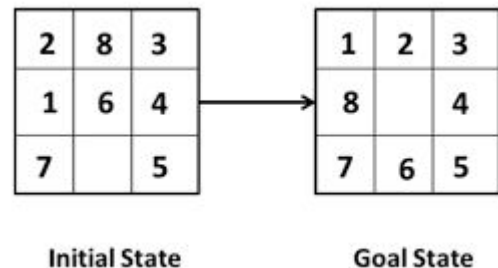
- The agent only has partial information about the state of the environment due to limitations in sensors or inherent characteristics of the environment.



Deterministic vs. Stochastic

Deterministic Environment:

- The next state of the environment is completely determined by the current state and the actions performed by the agent.
- There is no uncertainty in the outcome of actions.



Stochastic Environment:

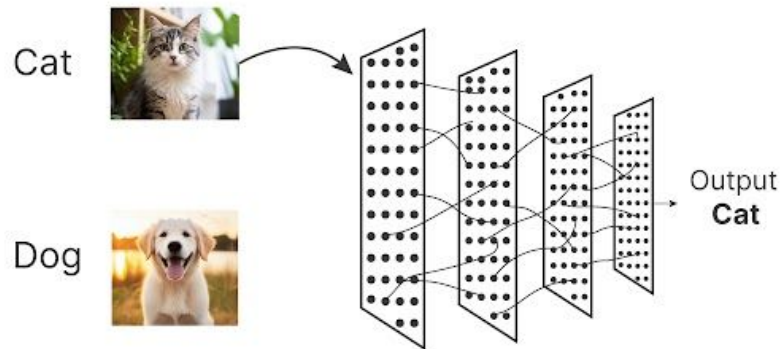
- The next state of the environment is not fully predictable due to random elements or inherent uncertainty.
- Actions may lead to different outcomes with certain probabilities.



Episodic vs. Sequential

Episodic Environment:

- The agent's actions are divided into discrete episodes.
- Each episode is independent of the others, and the outcome of one episode does not affect the next.



Sequential Environment:

- The current decision could affect all future decisions.
- The agent's actions are part of a continuous sequence, and each action influences subsequent actions.



Static vs. Dynamic

Static Environment:

- The environment remains unchanged while the agent is making a decision.
- The agent does not need to worry about the environment changing in the middle of its decision process.



Dynamic Environment:

- The environment can change while the agent is making a decision, requiring the agent to adapt to changes in real-time.



Discrete vs. Continuous

Discrete Environment:

- The environment consists of a finite number of distinct states and actions.
- The agent's decisions are based on a limited set of possibilities.



Continuous Environment:

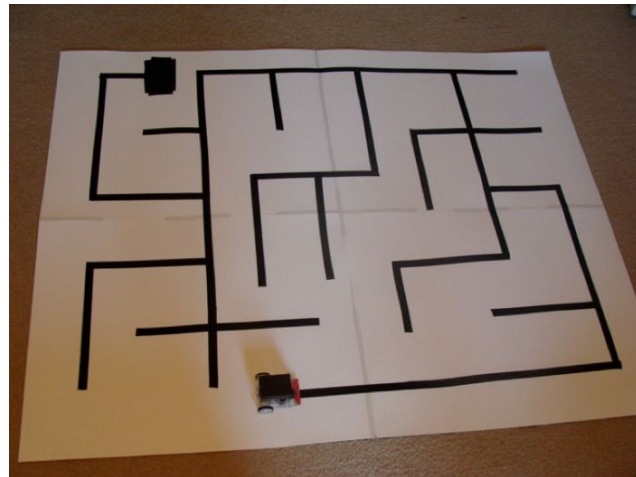
- The environment has a continuous range of states and actions.
- The agent must handle a potentially infinite number of possibilities.



Single-agent vs. Multi-agent

Single-agent Environment:

- Only one agent is operating in the environment,
- and there are no other agents to interact with or compete against.



Multi-agent Environment:

- Multiple agents operate within the environment, interacting with each other.
- These interactions can be competitive or cooperative.



Known vs. Unknown

Known Environment:

- The agent has complete knowledge of the environment's dynamics, including how actions affect the state of the environment.
- This allows for precise planning and decision-making.



Unknown Environment:

- The agent has no prior knowledge of the environment's dynamics
- and must learn how actions affect states through exploration and interaction.



Task Environment	Observable	Agents	Deterministic	Episodic	Static	Discrete
Crossword puzzle	Fully	Single	Deterministic	Sequential	Static	Discrete
Chess with a clock	Fully	Multi	Deterministic	Sequential	Semi	Discrete
Poker	Partially	Multi	Stochastic	Sequential	Static	Discrete
Backgammon	Fully	Multi	Stochastic	Sequential	Static	Discrete
Taxi driving	Partially	Multi	Stochastic	Sequential	Dynamic	Continuous
Medical diagnosis	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
Image analysis	Fully	Single	Deterministic	Episodic	Semi	Continuous
Part-picking robot	Partially	Single	Stochastic	Episodic	Dynamic	Continuous
Refinery controller	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
Interactive English tutor	Partially	Multi	Stochastic	Sequential	Dynamic	Discrete

Examples of task environments and their characteristics

Types of Agents

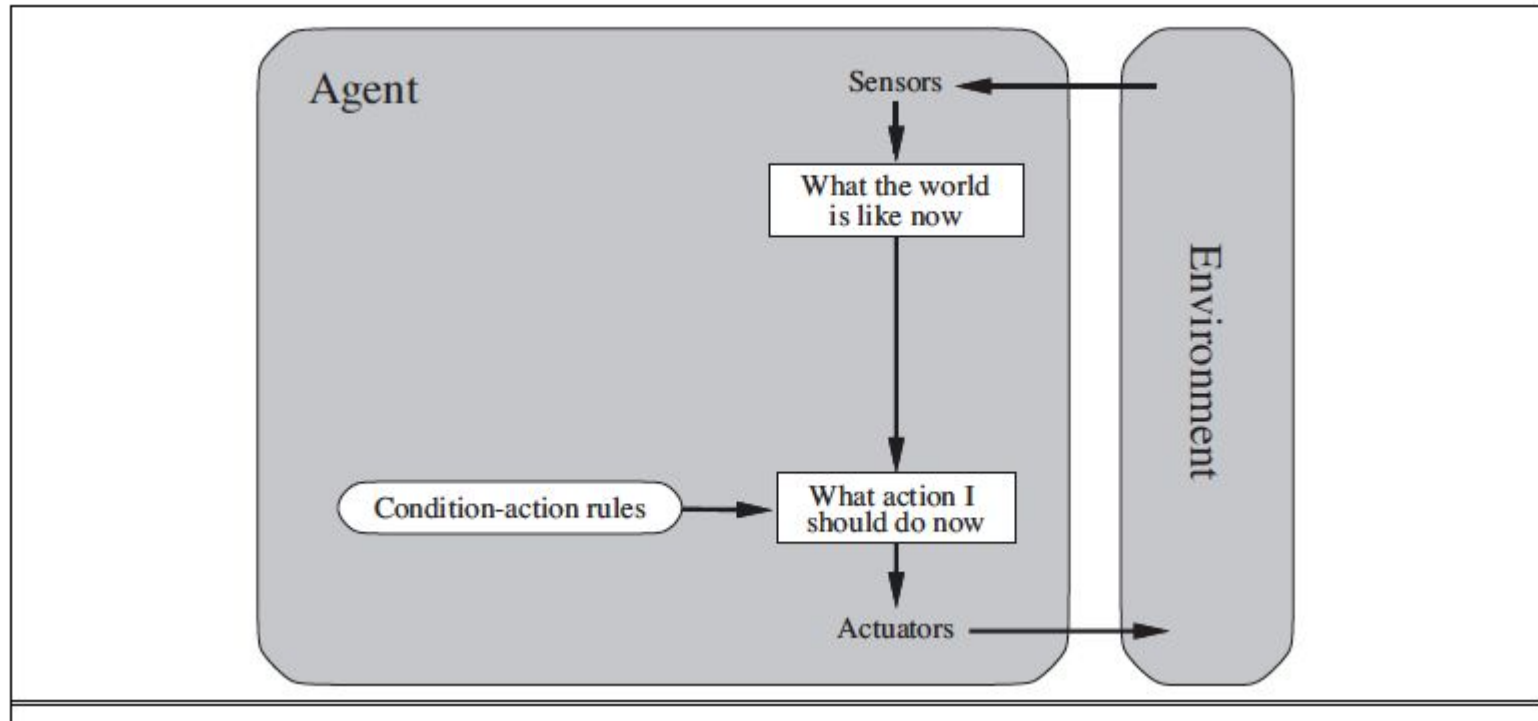
- **Simple Reflex Agents**
- **Model-Based Reflex Agents**
- **Goal-Based Agents**
- **Utility-Based Agents**

Simple Reflex Agents (SRA)

Simple Reflex Agents operate by selecting actions based solely on the current percept (i.e., what they sense at the moment), ignoring the rest of the percept history. They follow condition-action rules (if-then statements) to make decisions. They do not consider the history of percepts or the internal state of the environment.

How Simple Reflex Agents Work

- Perception: The agent perceives the environment through its sensors.
- Condition-Action Rules: The agent evaluates the current percept against a set of predefined rules.
- Action: Based on the matching condition, the agent performs the corresponding action.



Schematic diagram of a simple reflex agent.

SRA: Characteristics

1. **Direct Response:**

- Simple reflex agents respond directly to percepts with actions. They do not maintain any internal state or history of previous percepts.

2. **Condition-Action Rules:**

- The behavior of these agents is governed by a set of rules, often referred to as "if-then" rules. For example, "if the car in front is braking, then decelerate."

3. **No Memory:**

- These agents do not have memory; they do not keep track of past percepts or actions. Their decisions are made purely on the basis of the current percept.

4. **Limited Applicability:**

- Simple reflex agents are effective in environments that are fully observable and predictable. However, they are less effective in complex or partially observable environments where decisions need to consider past events or future consequences.

SRA: Strengths and Limitations

Strengths:

- **Simplicity:** Simple reflex agents are easy to design and implement due to their straightforward nature.
- **Efficiency:** In well-defined, predictable environments, they can operate efficiently without the need for complex processing.

Limitations:

- **Lack of Adaptability:** Without memory or consideration of the history, these agents cannot adapt to changes or learn from past experiences.
- **Limited Scope:** They are not suitable for complex environments where the current percept alone does not provide enough information to make an optimal decision.

SRA Example

Example: A basic thermostat:

- **Percept:** Current room temperature.
- **Condition-Action Rule:** If the temperature is below 20°C, turn on the heater; if the temperature is above 25°C, turn off the heater.
- **Action:** Turning the heater on or off based on the current temperature.

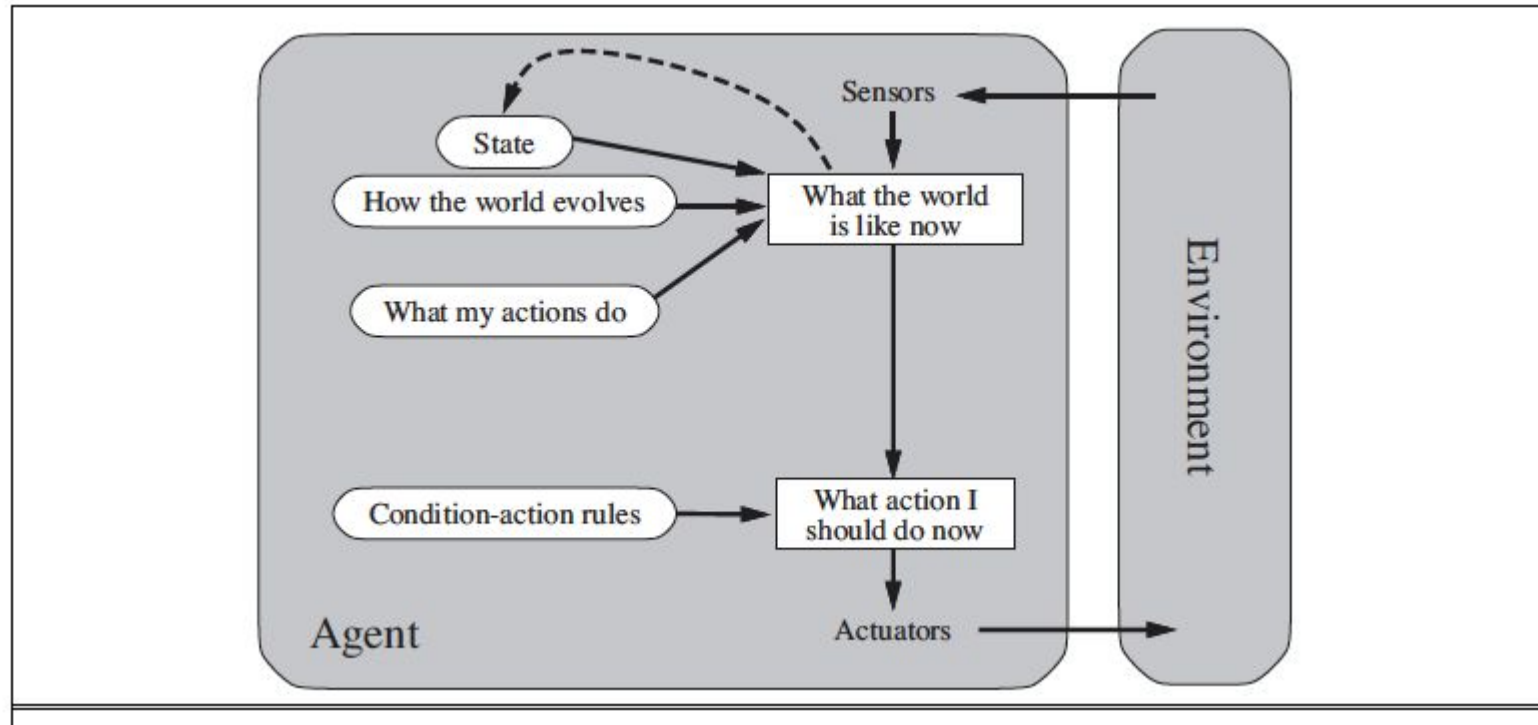


Model-Based Reflex Agents (MBRA)

Model-Based Reflex Agents maintain an internal state that depends on the percept history and a model of how the world works. This model helps the agent to keep track of parts of the world it can't see at the moment and to make decisions based on a combination of current percepts and stored information.

How Model-Based Reflex Agents Work

1. **Perception:** The agent perceives the environment through its sensors.
2. **Update State:** The agent updates its internal state based on the current percept and its internal model.
3. **Condition-Action Rules:** The agent evaluates its internal state and current percept against a set of predefined rules.
4. **Action:** Based on the matching condition, the agent performs the corresponding action.



Schematic diagram of a model-based reflex agent

MBRA: Characteristics

1. **Internal State:**

- Model-based reflex agents maintain an internal state that depends on the percept history. This internal state captures relevant aspects of the environment that the agent cannot directly perceive at any given time.

2. **World Model:**

- These agents use a model of the world that describes how the world evolves in response to actions and how percepts are generated. This model helps the agent update its internal state based on new percepts.

3. **Condition-Action Rules:**

- Like simple reflex agents, model-based reflex agents use condition-action rules. However, these rules are applied to the internal state rather than directly to the percepts.

4. **State Updating:**

- The agent updates its internal state using the model of the world whenever it receives a new percept. This process involves incorporating new information into the existing state.

MBRA: Strengths and Limitations

Strengths:

- **Memory:** The ability to maintain an internal state allows these agents to handle more complex and dynamic environments.
- **Increased Applicability:** Model-based reflex agents can function effectively in partially observable environments where not all relevant information is available at once.
- **Improved Decision Making:** By using an internal model, these agents can anticipate the outcomes of their actions better than simple reflex agents.

Limitations:

- **Complexity:** Maintaining and updating an internal state and world model increases the complexity of the agent.
- **Computation:** The need to update the internal state and apply condition-action rules to this state can require more computational resources.

MBRA: Example

Example: A robotic vacuum cleaner:

- **Percept:** Current position, obstacles detected.
- **Internal State:** Map of the environment, battery level, cleaning status.
- **Condition-Action Rule:** If the battery level is low and near the charging station, return to the charging station; if an obstacle is detected, change direction.
- **Action:** Moving to the charging station, changing direction, continuing cleaning.

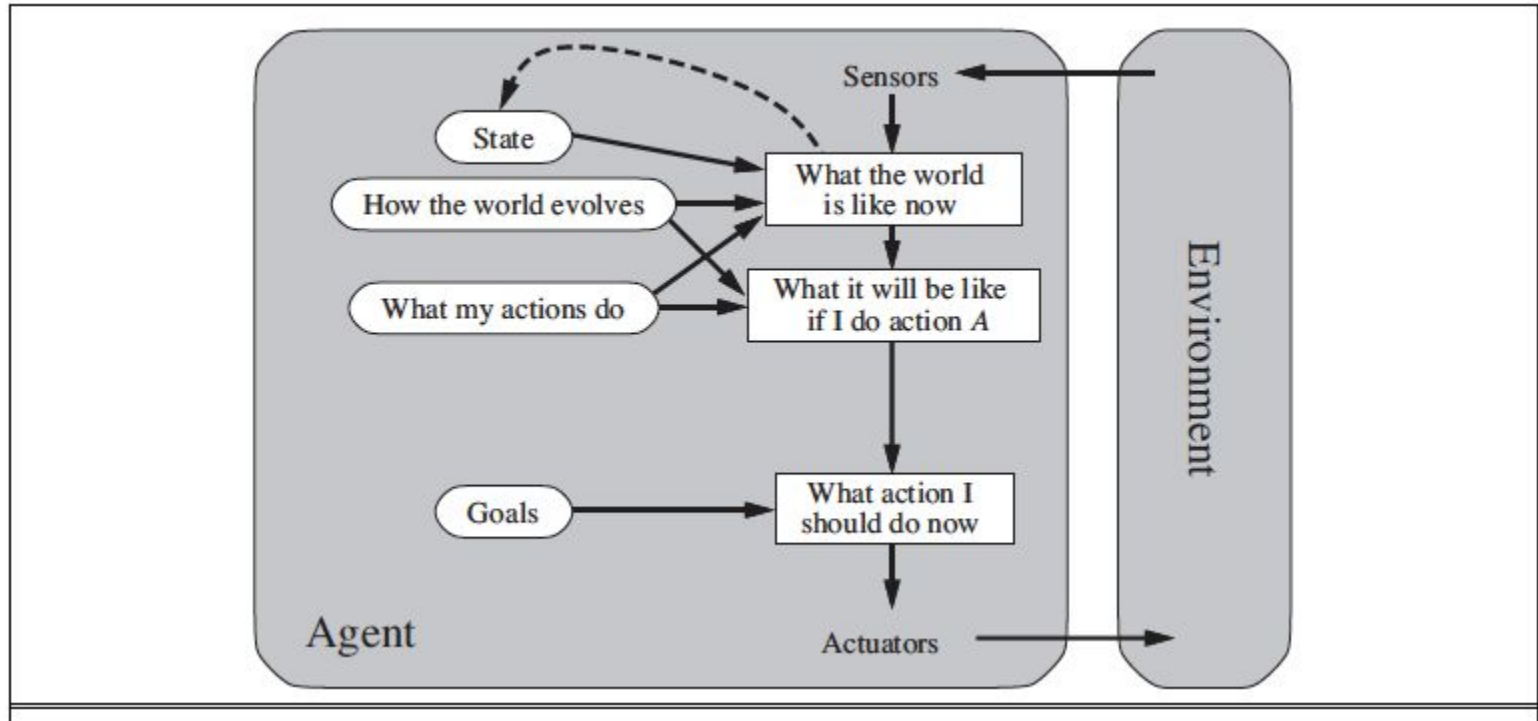


Goal-Based Agents (GBA)

Goal-Based Agents make decisions based on achieving specific goals. These agents evaluate different actions by considering how well each action will help them achieve their goals. They often use planning and search algorithms to find sequences of actions that lead to the desired outcome.

How Goal-Based Agents Work

1. **Perception:** The agent perceives the environment through its sensors.
2. **Goal Identification:** The agent has one or more goals to achieve.
3. **Planning:** The agent evaluates possible actions and sequences of actions to determine which ones will lead to achieving the goal.
4. **Action:** The agent performs the action that is most likely to achieve its goal.



A model-based, goal-based agent. It keeps track of the world state as well as a set of goals it is trying to achieve, and chooses an action that will (eventually) lead to the achievement of its goals.

GBA Characteristics

1. **Goals:**

- Goals are specific desired outcomes or states that the agent aims to achieve. These goals provide a purpose and direction for the agent's actions.

2. **Search and Planning:**

- Goal-based agents often employ search and planning algorithms to determine a sequence of actions that will lead them from their current state to the goal state. This involves exploring possible future states and evaluating the best path to the goal.

3. **Decision Making:**

- These agents consider not just the immediate effects of their actions but also their long-term consequences. This forward-thinking approach enables them to handle more complex tasks and environments.

4. **Adaptability:**

- Goal-based agents can adapt to changes in the environment by re-evaluating their plans and making new decisions based on updated information.

GBA: Strengths and Limitations

Strengths:

- **Effective in Complex Environments:** Goal-based agents can handle complex, dynamic environments where planning and foresight are necessary.
- **Purpose-Driven:** The use of goals provides clear criteria for success, enabling agents to focus their efforts effectively.
- **Flexible and Adaptive:** These agents can re-plan and adapt to changes in the environment, maintaining their focus on achieving their goals.

Limitations:

- **Computational Complexity:** Planning and search algorithms can be computationally intensive, especially in large or complex environments.
- **Requires Clear Goals:** The effectiveness of a goal-based agent depends on well-defined, achievable goals. Ambiguous or conflicting goals can hinder performance.

GBA Example

Example: A GPS Navigation System

- **Perception:** Current location, traffic data, map data.
- **Goal Identification:** Destination address.
- **Planning:** Calculating the optimal route considering current traffic conditions.
- **Action:** Providing turn-by-turn directions to follow the optimal route.

Use Cases:

- **Robotics:** A warehouse robot tasked with retrieving an item and delivering it to a specific location.
- **Game AI:** Characters in video games that need to achieve specific objectives, such as capturing a flag or completing a mission.
- **Autonomous Vehicles:** Cars that plan routes to reach destinations efficiently.

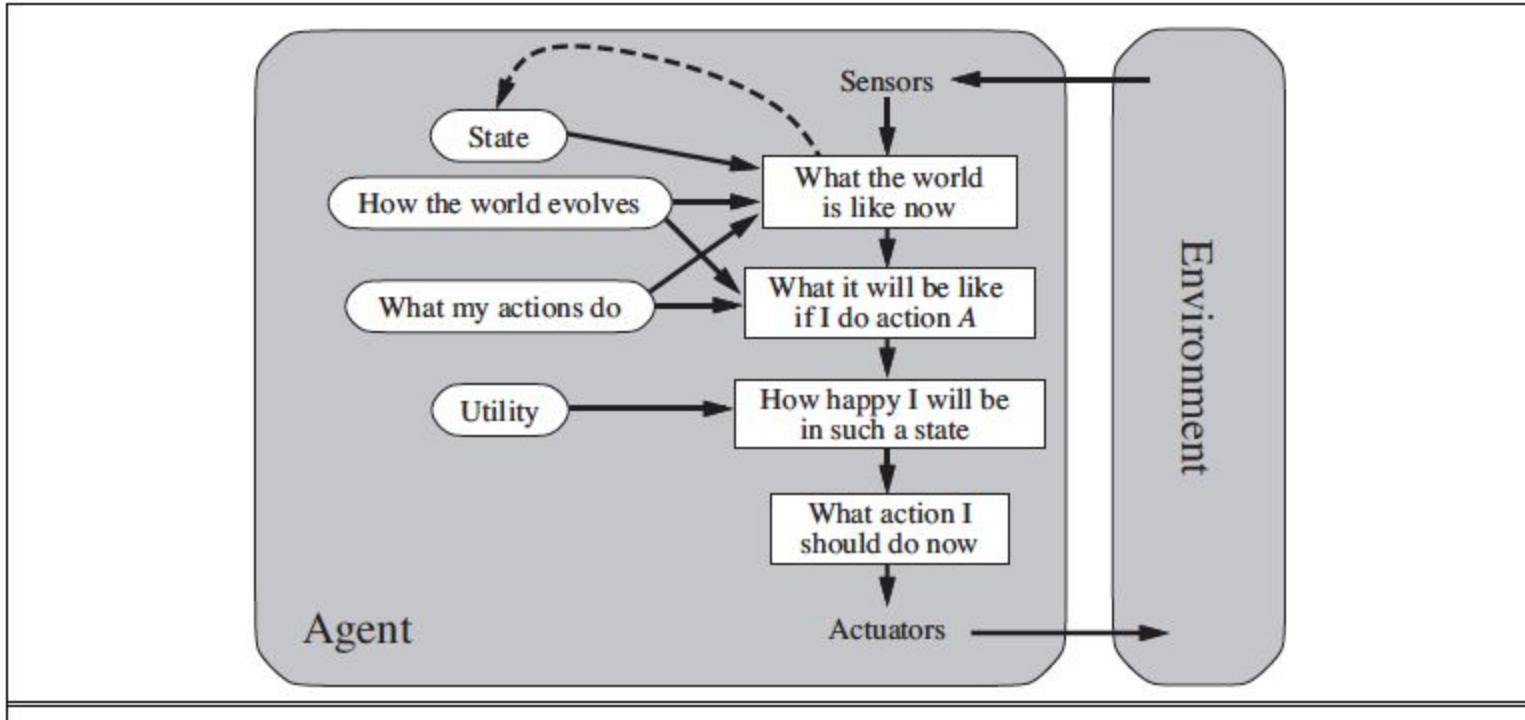


Utility-Based Agents (UBA)

Utility-Based Agents extend goal-based agents by considering not only whether goals are achieved but also the desirability (utility) of different states. They use a utility function to evaluate the desirability of different states or outcomes and choose actions that maximize their expected utility.

How Utility-Based Agents Work

1. **Perception:** The agent perceives the environment through its sensors.
2. **Utility Calculation:** The agent uses a utility function to evaluate the desirability of different states or outcomes.
3. **Decision Making:** The agent evaluates possible actions and chooses the one that maximizes expected utility.
4. **Action:** The agent performs the action that maximizes its utility.



A model-based, utility-based agent. It uses a model of the world, along with a utility function that measures its preferences among states of the world. Then it chooses the action that leads to the best expected utility, where expected utility is computed by averaging over all possible outcome states, weighted by the probability of the outcome.

UBA: Characteristics

1. **Utility Function:**

- A utility function is a mathematical representation that assigns a numerical value (utility) to each possible outcome. This value reflects the agent's preference for that outcome.

2. **Decision Making:**

- Utility-based agents use the utility function to evaluate and compare different outcomes, choosing the action that maximizes the expected utility. This involves considering not only the likelihood of different outcomes but also their desirability.

3. **Handling Trade-offs:**

- These agents can handle trade-offs between conflicting goals or objectives. For example, an agent might balance the trade-off between speed and safety in a navigation task.

4. **Risk Management:**

- Utility-based agents can manage uncertainty and risk by considering the expected utility of actions, which accounts for both the probability and the utility of different outcomes.

UBA: Strengths and Limitations

Strengths:

- **Optimal Decision Making:** Utility-based agents aim to make optimal decisions by maximizing the overall utility, leading to better outcomes.
- **Flexibility:** These agents can adapt to different situations and trade-offs by adjusting their utility function.
- **Handling Complex Environments:** Utility-based agents are well-suited for complex, dynamic environments where multiple factors need to be considered.

Limitations:

- **Complexity:** Defining and computing the utility function can be complex, especially in environments with many variables and uncertainties.
- **Subjectivity:** The utility function is subjective and must be carefully designed to reflect the agent's preferences accurately.

UBA: Example

Example: An Automated Trading System

- **Perception:** Market data, historical trends, economic indicators.
- **Utility Calculation:** Evaluating potential trades based on expected return and risk.
- **Decision Making:** Choosing trades that maximize expected utility (profit adjusted for risk).
- **Action:** Executing buy or sell orders.

Use Cases:

- **Healthcare:** Systems that recommend treatments by balancing efficacy, side effects, and patient preferences.
- **Finance:** Investment algorithms that choose portfolios by balancing expected returns and risks.
- **Resource Management:** Systems that allocate resources (like compute power in a data center) to maximize overall efficiency and performance.



Learning Agents

Learning Agents are capable of improving their performance over time by learning from their experiences. They adapt their behavior based on past outcomes, feedback, and interactions with the environment. Learning agents are designed to handle dynamic and complex environments where predefined rules might not be sufficient.

Key Components of Learning Agents

1. **Learning Element:** Responsible for making improvements and acquiring new knowledge or skills based on experience.
2. **Performance Element:** Executes actions based on the knowledge it has acquired. This component is similar to the decision-making part of non-learning agents.
3. **Critic:** Provides feedback to the learning element about the agent's performance. It evaluates the actions taken and helps the learning element to make adjustments.
4. **Problem Generator:** Suggests actions that lead to new experiences and opportunities for learning. It helps the agent explore new possibilities and gather diverse experiences.

Learning Agents

How Components Contribute to Learning

1. Learning Element:

- **Function:** Updates the agent's knowledge base or policies based on feedback and new data.
- **Contribution:** Enables the agent to adapt and improve by refining its understanding of the environment and the consequences of its actions.

2. Performance Element:

- **Function:** Uses the current knowledge to make decisions and take actions in the environment.
- **Contribution:** Acts as the operational part of the agent, implementing the learned strategies and behaviors.

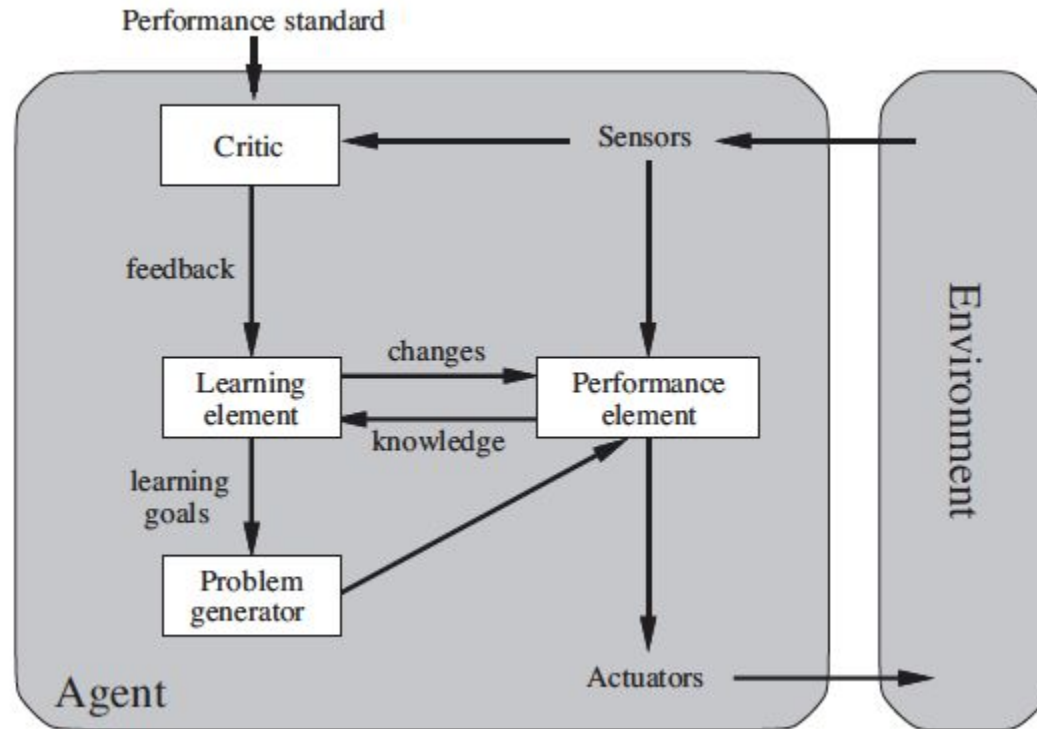
Learning Agents

3. Critic:

- **Function:** Evaluates the outcomes of the agent's actions and provides feedback.
- **Contribution:** Helps the learning element understand the effectiveness of different actions and identify areas for improvement.

4. Problem Generator:

- **Function:** Encourages exploration by suggesting new actions or strategies that the agent has not tried before.
- **Contribution:** Promotes learning by exposing the agent to new experiences and challenges, preventing it from getting stuck in suboptimal behaviors.



A general learning agent

Types of Learning in Agents:

Supervised Learning:

- In supervised learning, the agent is provided with labeled training data that includes input-output pairs. The agent learns to map inputs to outputs based on this data, refining its model to improve accuracy.
- Example: A learning agent that recognizes handwritten digits might be trained on a dataset of images labeled with the correct digit.

Unsupervised Learning:

- Unsupervised learning involves training the agent on data without explicit labels. The agent must identify patterns, clusters, or structures within the data on its own.
- Example: A learning agent tasked with customer segmentation might analyze purchase data to identify groups of similar customers without predefined categories.

Types of Learning in Agents:

Reinforcement Learning:

- In reinforcement learning, the agent learns by interacting with its environment and receiving feedback in the form of rewards or punishments. The goal is to learn a policy that maximizes cumulative rewards over time.
- Example: A robot navigating a maze might receive a reward for reaching the goal and penalties for hitting walls, learning to optimize its path to the goal.

Online Learning:

- Online learning refers to the process where the agent learns incrementally as new data arrives, updating its model continuously rather than relying on a fixed dataset.
- Example: A recommendation system that adapts in real-time based on users' interactions with recommended content.

Examples of Learning Agents

1. Supervised Learning Agents

Example: Spam Email Classifier

- **Description:** A spam email classifier learns to differentiate between spam and non-spam emails based on labeled training data.
- **Components:**
 - **Learning Element:** Uses labeled examples to update its classification model.
 - **Performance Element:** Classifies incoming emails based on the learned model.
 - **Critic:** Evaluates the classifier's accuracy using feedback from users (e.g., marking emails as spam or not spam).
 - **Problem Generator:** Incorporates new types of emails into the training set to improve classification.

Examples of Learning Agents

2. Unsupervised Learning Agents

Example: Customer Segmentation

- **Description:** A marketing agent that segments customers into different groups based on purchasing behavior without labeled data.
- **Components:**
 - **Learning Element:** Uses clustering algorithms to group customers based on similarities in data.
 - **Performance Element:** Applies the learned segments to target marketing campaigns.
 - **Critic:** Analyzes the success of marketing campaigns to refine the segmentation.
 - **Problem Generator:** Identifies new customer attributes to consider for segmentation.

Examples of Learning Agents

3. Reinforcement Learning Agents

Example: AlphaGo

- **Description:** AlphaGo, developed by DeepMind, uses reinforcement learning to play the game of Go. It learns optimal strategies by playing numerous games against itself and other opponents, constantly improving based on the outcomes.
- **Components:**
 - **Learning Element:** Updates its policy based on the results of games played.
 - **Performance Element:** Executes moves during gameplay.
 - **Critic:** Evaluates the outcome of each game to determine the success of different strategies.
 - **Problem Generator:** Explores new strategies and move sequences to enhance learning.

Challenges and Considerations

Exploration vs. Exploitation: Learning agents must balance exploration (trying new actions to gather more information) and exploitation (using known information to maximize rewards). This balance is critical to avoid local optima and ensure robust learning.

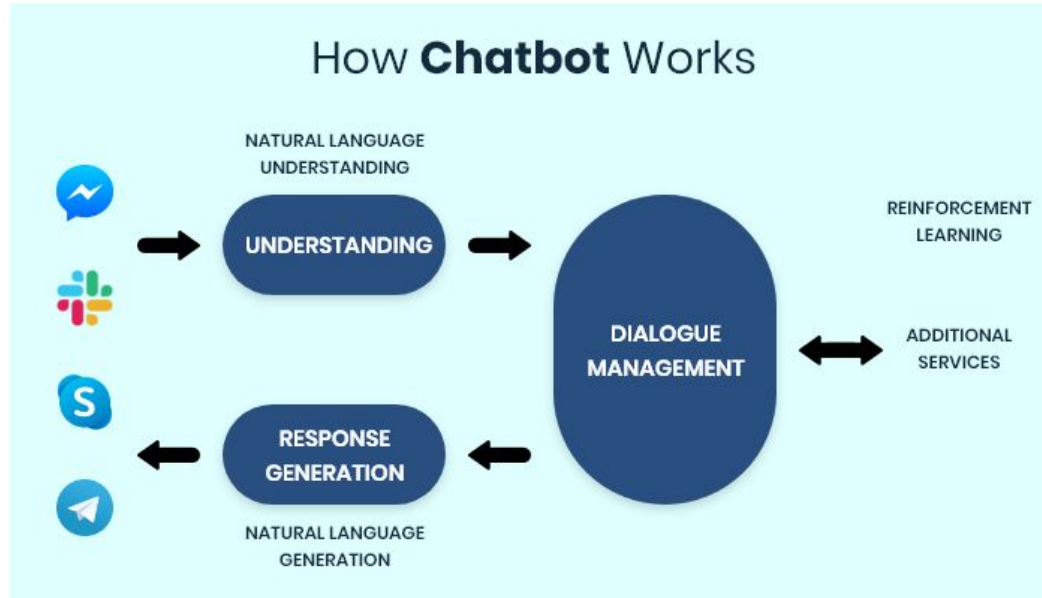
Scalability: As the complexity of the environment increases, the agent must handle larger state and action spaces, which can pose computational challenges.

Transfer Learning: Learning agents may need to transfer knowledge from one domain or task to another, requiring them to generalize and apply what they've learned in new contexts.

Ethical Considerations: In some applications, learning agents must be designed with ethical considerations in mind, especially when their decisions impact human well-being, such as in healthcare or autonomous vehicles.

Case Study: Virtual Personal Assistants

- **Learning Element:** The learning element in a chatbot is responsible for improving the chatbot's ability to understand and generate responses.
- **Performance Element:** The performance element involves the actual operation of the chatbot during interactions with users. It selects responses based on the current state of the conversation
- **Critic:** The critic evaluates how well the chatbot performs in its interactions. (User feedback, sentiment analysis)
- **Problem Generator:** The problem generator in a chatbot might involve testing new types of interactions or exploring different conversation paths to expand the chatbot capabilities.



Other Examples of Intelligent Agents

- **Smart Home Systems:** Systems like Nest smart thermostats and Philips Hue smart lighting adapt to user preferences and environmental conditions. They learn from user behavior to optimize energy usage and provide comfort.
- **Robotics:** Robots in manufacturing, healthcare, and service industries perform tasks autonomously or semi-autonomously. For instance, robotic arms in factories assemble products, and robotic surgery systems assist surgeons in performing precise operations.
- **Game AI:** Non-player characters (NPCs) in video games exhibit intelligent behavior, such as planning, learning, and adapting to player actions. Examples include AI opponents in strategy games like StarCraft or role-playing games like Skyrim.
- **E-commerce Recommendations:** Online retailers like Amazon use intelligent agents to recommend products to users based on their browsing and purchase history. These agents analyze large datasets to identify patterns and suggest items that users are likely to be interested in.